Grade 5 Natural Sciences and Technology: Processing materials and fair testing

Natural Sciences and Technology · Grade 5 · 30 marks · 60 minutes

Learner name: _____ Date: _____

Time: 60 minutes Total: 30 marks

Instructions

- Answer all five questions.
- Show calculations where requested.
- Use evidence from the printed stimulus when a question refers to it.

Assessment

Read the stimulus and each question carefully.

> *A Grade 5 CAPS example describes mixing sand, gravel, cement and water to make concrete. Learners will prepare two small concrete samples with different amounts of gravel. They will keep the other materials, moulds, setting time and test method the same. After the samples set, an educator will add identical counters one at a time and record how many each sample supports before it cracks. The class will also record material properties before mixing and after the samples set.*

1.1 Name the four materials mixed in the example and name the processed material they make. **[5]**

1.2 State what property information learners should record before mixing and after the samples set. Give two examples of properties to compare and explain why they record both stages. **[5]**

1.3 In the two-sample test, identify what is changed, what is measured and three conditions that should be kept the same. Explain why this makes the comparison fair. **[6]**

1.4 Based on the planned test, predict which sample will support more counters, or predict that they will support the same number. Explain why your prediction is reasonable, state what result will count as evidence, and give one reason the result should be checked with repeated trials. **[7]**

1.5 Give suitable results-table headings, including units where needed, for the gravel test. Then describe one way to keep the comparison reliable and one limitation of using small samples to choose material for a real path. **[7]**

Source basis: DBE CAPS Natural Sciences and Technology · intermediate_grade_4_6. See the grade receipt for the official PDF URL and SHA-256.

DRAFT — Qualified educator review required before classroom use.